

Exciton torque

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Useful references and info about the exciton in tj model

I. USEFUL REFERENCES

[1] They compute exciton spectra and optical conductivity using t-J model on a four-leg cylinder using MPS (Their model is the typical t-j model with Hubbard and J between nearest neighbors), starting from half-filling ground state and applying an exciton operator, where, depending on the dopant (fermionic, bosonic, or a mixture) the corresponding excitonic operator is defined. More details about the MPS simulations are in Appendix C. Their ultimate goal is to compare against the Fermi-Hubbard and understand the nature of their excitations, e.g., exciton vs distinguishable dopants, allowing the tracking of the nature of the Fermi-Hubbard excitons.

II. MODEL AND METHOD

We consider an effective electron-hole model with spin degrees of freedom coupled to localized semiclassical spins

$$\hat{H} = \hat{H}_0 + \hat{H}_{sd}[\mathbf{M}(t)] + \hat{H}_U + \hat{H}_V \quad (1)$$

$$\hat{H}_0 = \sum_{\langle ij \rangle, l} \hat{c}_i^{\dagger l} H_{0,ij}^l \hat{c}_j^l \quad (2)$$

$$\hat{H}_U = U \sum_{il} \hat{n}_{i\uparrow}^l \hat{n}_{i\downarrow}^l \quad (3)$$

$$\hat{H}_V = -V \sum_{\sigma\sigma' l} \hat{n}_{i\sigma}^l \hat{n}_{i\sigma'}^l \quad (4)$$

$$\hat{H}_{sd}(t) = -J_{sd} \sum_{i,l} \mathbf{M}_i^l(t) \cdot \hat{\mathbf{s}}_i^l \quad (5)$$

The standard semiclassical dynamics for the LMMs coupled to electrons is obtained as a saddle point of the Gaussian approximation of the effective action (once the electrons are integrated out), which only works at small coupling $j_{sd} \ll \gamma$, obtaining an effective nonlocal damping. In this approximation, the Kernel is obtained from the magnetic susceptibility of the electrons:

$$R_{ij}^{ab}(t) = J_{sd}^2 \int_{-\infty}^t dt' \chi_{ij}^{ab,R}(t-t') \quad (6)$$

$$\chi_{ij}^{ab,R}(t, t') = -i\theta(t-t') \langle [\hat{s}_i^a(t), \hat{s}_i^b(t')] \rangle \quad (7)$$

Thus by exact methods, it is possible to get full susceptibility. On the other hand, once the excitonic effects are included...

The bare dynamics of $\mathbf{M}_i^l(t)$ is described by the classical hamiltonian $H_{cl}[\mathbf{M}]$. Our goal is to derive an effective semiclassical dynamics for the LMMs, including the effects of correlated electrons within each layer and exciton formation in between layers.

$$\partial_t \mathbf{M}^l(t) = \mathbf{M}^l(t) \times [\mathbf{B}_{cl}^l[\{\mathbf{M}(t)\}] + \mathbf{B}_{\Phi}^l[\{\mathbf{M}(t)\}] + \mathbf{B}_0^l[\{\mathbf{M}(t)\}]] \quad (8)$$

$$\mathbf{B}^{l,cl}[\mathbf{M}(t)] = -\frac{\partial H^{cl}[\{\mathbf{M}(t)\}]}{\partial \mathbf{M}^l(t)} \quad (9)$$

A. Excitonic torque ($U = 0$)

We write the effective action as a function of the intralayer excitonic order parameter $\hat{B}_i^{\sigma\sigma'} = \hat{c}_{i\sigma}^{\dagger} \hat{c}_{i\sigma'}^2$, using $\hat{n}_i^1 \hat{n}_i^2 = 2\hat{n}_i^1 - \sum_{\sigma,\sigma'} \hat{B}_i^{\sigma\sigma'} \hat{B}_i^{\sigma\sigma'}$ and absorbing the first term into the chemical potential, the electronic action $\mathcal{S}_e$ reads

$$\mathcal{S}_e = \int_{\mathcal{C}} dz dz' \sum_{ij} \bar{\Psi}_i^T G_{\mathbf{M}}^{-1} \Psi_j + V \int_{\mathcal{C}} dz \sum_{i\nu} B_i^{\nu} \bar{B}_i^{\nu}, \quad (10)$$

where $\Psi = (c_{\uparrow}^1, c_{\downarrow}^1, c_{\uparrow}^2, c_{\downarrow}^2)^T$, and the polarized Green function (GF) satisfies $(i\partial_z - H_0 - H_{sd}[\mathbf{M}])G_{\mathbf{M}}(z, z') = \delta_C(z, z')$. We additionally decompose the excitonic field on its charge and spin degrees of freedom using $B_i^{\nu} = \text{Tr}_{\sigma}[\Gamma^{\nu} B_i]$, with $\Gamma^{\nu} = \sigma^{\nu}/\sqrt{2}$ and $\nu = 0, x, y, z$. Applying the Hubbard Strattonovich (HS) transformation $\mathcal{S}_V \rightarrow \mathcal{S}_V^{HS}[\{\Phi_i^{\nu}\}]$, the action is written as

$$\begin{aligned} \mathcal{S}_e = & \int_{\mathcal{C}} dz dz' \sum_{ij} \bar{\Psi}_i^T (G_{\mathbf{M},ij}^{-1} - \delta_{ij} \mathcal{V}_{\Phi})_{ij} \Psi_j \\ & + \frac{1}{V} \int_{\mathcal{C}} dz \sum_{i,\nu} \bar{\Phi}_i^{\nu} \Phi_i^{\nu}. \end{aligned} \quad (11)$$

Integrating out the electrons and keeping to second order (RPA), the effective exciton action reads

$$\mathcal{S}_\Phi^{\text{eff}}[\mathbf{M}] = \int_c dz dz' \sum_{\nu\nu'} \bar{\Phi}^\nu D_{\nu\nu'}^{-1} \Phi^{\nu'} - i \text{Tr} \ln[G_{\mathbf{M}}^{-1}] \quad (12)$$

where we assumed a homogeneous order parameter for the exciton, the propagator of the exciton satisfies the Bethe-Salpeter equation $D = V(1 + \Pi D)$

$$D_{\mu\nu}^{-1}(z, z') = \frac{\delta_C(z, z') \delta_{\mu\nu}}{V} - \Pi_{\mu\nu}(z, z') \quad (13)$$

$$\Pi_{\mu\nu}(z, z') = -\frac{i}{L} \text{Tr}[\Gamma^{\mu\dagger} G_{\mathbf{M}}^1(z, z') \Gamma^\nu G_{\mathbf{M}}^2(z', z)], \quad (14)$$

Additionally, we integrate out the exciton degrees of freedom to get an effective action for the classical moments. The total effective action reads,

$$\mathcal{S}_{\text{cl}}^{\text{eff}} = \mathcal{S}_{\text{cl}} - i \text{Tr} \ln[G_{\mathbf{M}}^{-1}] + i \text{Tr} \ln[D^{-1}] \quad (15)$$

In order to find the semiclassical dynamics, we minimize with respect to the classical LLMs $\frac{\partial \mathcal{S}_{\text{cl}}^{\text{eff}}}{\partial \mathbf{M}^l} = 0$, the

classical term, which contains the Berry phase and its respective classical Hamiltonian of the LLMs returns

$$\frac{\partial \mathcal{S}_{\text{cl}}}{\partial \mathbf{M}^l(t)} = \frac{1}{\gamma_l} \mathbf{M}^l \times \partial_t \mathbf{M}^l + \mathbf{B}^{l, \text{cl}} \quad (16)$$

The contributions to the effective field from the polarized electrons is

$$\begin{aligned} \mathbf{B}_0^l[\{\mathbf{M}(t)\}] &= i \frac{\partial \text{Tr} \ln[G_{\mathbf{M}}^{-1}]}{\partial \mathbf{M}^l} \\ &= J_{sd} \langle \hat{\mathbf{S}}^l \rangle_{\mathbf{M}} \end{aligned} \quad (17)$$

and the contribution from the exciton is

$$\begin{aligned} \mathbf{B}_\Phi^l[\{\mathbf{M}(t)\}] &= -i \frac{\partial \text{Tr} \ln[D^{-1}]}{\partial \mathbf{M}^l} \\ &= -i \text{Tr} \left[D \frac{\partial \Pi}{\partial \mathbf{M}^l} \right] \end{aligned} \quad (18)$$

Where the explicit form of the torques is: (To compute the real-time dynamics, we should apply Langreth rules to get the retarded component):

$$\begin{aligned} B_{\alpha\Phi}^1(\bar{z}) &= \int dz dz' \frac{j_{sd}}{L} \sum_{\mu\nu} D_{\mu\nu}(z, z') \text{Tr}[\Gamma^\mu G_{\mathbf{M}}^2(z', \bar{z}) \Gamma^\nu G_{\mathbf{M}}^1(\bar{z}, z') \sigma^\alpha G_{\mathbf{M}}^1(z', z)] \\ B_{\alpha\Phi}^2(\bar{z}) &= \int dz dz' \frac{j_{sd}}{L} \sum_{\mu\nu} D_{\mu\nu}(z, z') \text{Tr}[\Gamma^\mu G_{\mathbf{M}}^2(z', \bar{z}) \sigma^\alpha G_{\mathbf{M}}^2(\bar{z}, z) \Gamma^\nu G_{\mathbf{M}}^1(z, z')] \end{aligned} \quad (19)$$

Putting all together, and following the standard procedures, we finally arrive to Eq. [(??)]

Up to this point, the theory is completely self-consistent, $\mathbf{M} \rightarrow G_{\mathbf{M}} \rightarrow D \rightarrow \mathbf{M}$. At each time step, the classical magnetization, the polarized GF, and the exciton propagator should be evolved self-consistently. There are two paths that can be followed:

- Given a classical trajectory of the LLMs, calculate the exciton response and spectra from $D[\mathbf{M}]$
- Use a j_{sd} expansion to get explicit expressions for torques without self-consistency

1. j_{sd} expansion and LLG equations

In order to get the standard LLG, the effective fields should be expanded to first order in $\mathbf{M}$. This can be done in two different ways:

- Considering an adiabatic approach, where $G_{\mathbf{M}}$ is computed explicitly from a frozen configuration.

(It sounds like Born-Oppenheimer nucleus-electron motion), and still evolve self-consistently

- Expanding to first order $G_{\mathbf{M}} = j_{sd} \mathbf{M} G^{(1)} + O(j_{sd}^2)$ and eliminate the selfconsistency because the kernels can be computed independently

The second way emerges naturally as a second-order expansion of S^{eff} , and is the natural way to get the effective LLG equation type. In our case, the susceptibility comes from

$$(j_{sd})^2 \chi_{ab}^{ll'}(z, z') = \frac{\partial \mathcal{S}^{\text{eff}}}{\partial M_a^l(z) \partial M_b^{l'}(z')} \quad (20)$$

For the electronic mediated kernel, the standard non-local kernel [2], where only the retarded component contributes to the equations

$$R_{ab}^{0, ll'}(z, z') = (j_{sd})^2 \delta_{ll'} \sum_k \text{Tr}[\sigma^a g_k^l(z, z') \sigma^b g_k^l(z', z)] \quad (21)$$

$$= (j_{sd})^2 \delta_{ll'} \delta_{ab} \sum_k g_k^l(z, z') g_k^l(z', z) \quad (22)$$

The susceptibility contribution from the exciton is:

$$R_{ab}^{\Phi(I),ll'}(z, z') = -i\text{Tr}_\nu[D^0 \cdot \Lambda_{ab}^{ll'}(z, z')] \quad (23)$$

$$R_{ab}^{\Phi(II),ll'}(z, z') = i\text{Tr}_\nu[D^0 \cdot \Lambda_l^a(z) \cdot D^0 \cdot \Lambda_{l'}^b(z')](24)$$

where the vertex functions $\Lambda_a^l(z, z'; \bar{z}) = \frac{\partial \Pi_\mu(z, z')}{\partial M_a^l(\bar{z})}$ are:

$$\Lambda_a^1(z, z'; \bar{z}) = i \frac{J_{sd}}{L} \sum_k \text{Tr}[\Gamma^\mu g^2(z, z') \Gamma^\nu g^1(z', \bar{z}) \sigma^a g^1(\bar{z}, z)]$$

$$\Lambda_a^2(z, z'; \bar{z}) = i \frac{J_{sd}}{L} \sum_k \text{Tr}[\Gamma^\mu g^2(z, z') \sigma^a g^2(z', \bar{z}) \Gamma^\nu g^1(\bar{z}, z)]$$

$\Lambda_a^l(z')$ denotes the 1 and 2 vertex

2. Adiabatic expansion, damping, inertia and so ...

To derive explicit non-local...

B. Semiclassical Low U Effective Theory ($U, V \ll \gamma$)

$$S_V = V \int_{\mathcal{C}} dz \sum_{i\sigma\sigma'} B_i^{\sigma\sigma'} \bar{B}_i^{\sigma\sigma'} \quad (25)$$

C. Semiclassical Strong U Effective Theory or Bilayer tJ Model ($U \gg \gamma$)

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- [1] A. Bohrdt, E. Demler, and F. Grusdt, Spectroscopy of hubbard-mott excitons and their ro-vibrational excitations, arXiv preprint arXiv:2406.16854 (2024).
 - [2] F. Reyes-Osorio and B. K. Nikolić, Gilbert damping in metallic ferromagnets from schwinger-keldysh field theory:

Intrinsically nonlocal, nonuniform, and made anisotropic by spin-orbit coupling, [Phys. Rev. B **109**, 024413 \(2024\)](#).